

Project Design Phase

Solution Architecture

Date	27/09/2026
Team ID	SWTID-2026-8900
Project Name	Streamlining IT Procurement – Automating Standard Laptop Orders with Flow Designer

Solution architecture describes how the proposed system connects the user, IT service platform, approval process, procurement team, and automation workflow to solve the problem of manual laptop ordering.

Goals of the Solution Architecture

- Automate the standard laptop procurement process.
- Reduce manual activities and processing time.
- Provide an organized approval mechanism.
- Automatically notify employees and IT teams about request updates.
- Improve visibility and tracking of laptop requests.
- Standardize the procurement workflow.
- Make the solution scalable for other IT assets and services.

Solution Architecture Diagram

You can use the following architecture for your project:

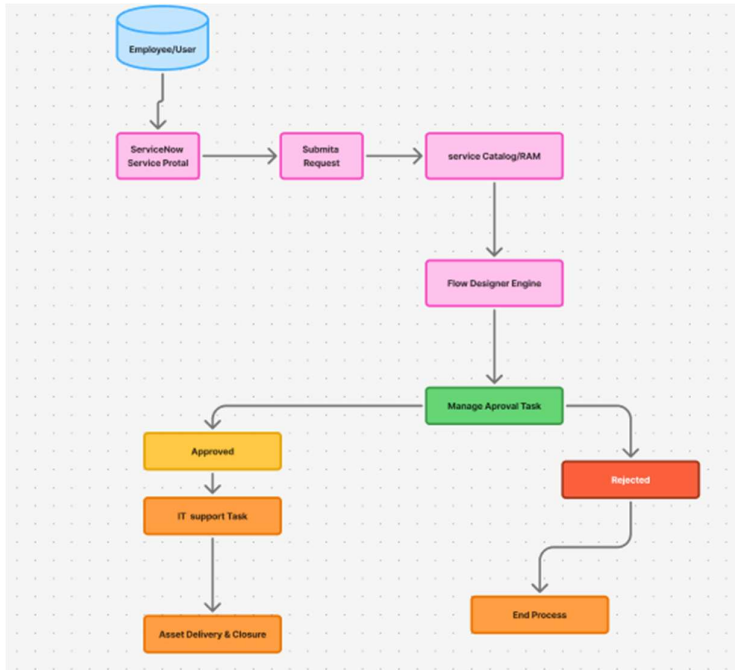


Figure 1: Solution Architecture and Data Flow for Automated Standard Laptop Procurement

Reference link:

<https://www.figma.com/board/XJlDhCFk76zDXd4jy0sn34/Untitled?node-id=0-1&t=3teSFRaakQZZuDub-1>

Architecture Components

Component	Purpose
Employee	Submits a request for a standard laptop.
ServiceNow / IT Request Portal	Provides the interface for submitting and tracking laptop requests.
Flow Designer	Automates the workflow and connects different stages of the procurement process.
Request Validation	Checks whether the submitted request contains the required information.
Manager Approval	Provides authorization before the laptop order is processed.
IT / Procurement Team	Processes the approved laptop request and manages fulfilment.

Component	Purpose
Request/Asset Records	Stores request details and status information.
Notification System	Automatically informs employees and relevant teams about approvals, rejection, and order progress.
Laptop Fulfilment	Represents the final stage where the approved laptop is prepared and delivered to the employee.

Data Flow

The architecture follows a simple automated flow:

Employee → ServiceNow Portal → Service Catalog → Flow Designer → Approval → IT Support → Asset Delivery → Request Closure

This architecture reduces dependency on manual emails and repeated communication. **Flow Designer acts as the central automation layer**, coordinating the different stages of the standard laptop procurement process.

Expected Benefits

The proposed architecture provides:

- Faster laptop request processing
- Automated approval routing
- Reduced manual intervention
- Improved request tracking
- Automatic notifications
- Fewer human errors
- Standardized procurement procedures

- Better employee experience
- Easier expansion to other IT asset requests

Reference: The architecture can be presented in the same style as the AWS architecture example provided in your template, while the components and data flow are customized specifically for your **Flow Designer-based laptop procurement project**.